Exploratory microbial genomics

MMB-127

Schedule

Week 1: Basics of UNIX and working with the command line
Data trimming
Genome assembly

Week 2: Genome annotation
Reference genomes

Week 3: Comparative genomics
Pangenomics
Wrap-up

Learning outcomes

After completing this module, you will be able to:

- Choose the most adequate platform for your genome sequencing experiment
- Investigate and judge the quality of sequencing data
- Make use of a variety of tools to:
  - Process whole genome sequencing data
  - Assemble and annotate whole genome sequencing data
  - Predict metabolic pathways from assembled and annotated genomes

Practical things

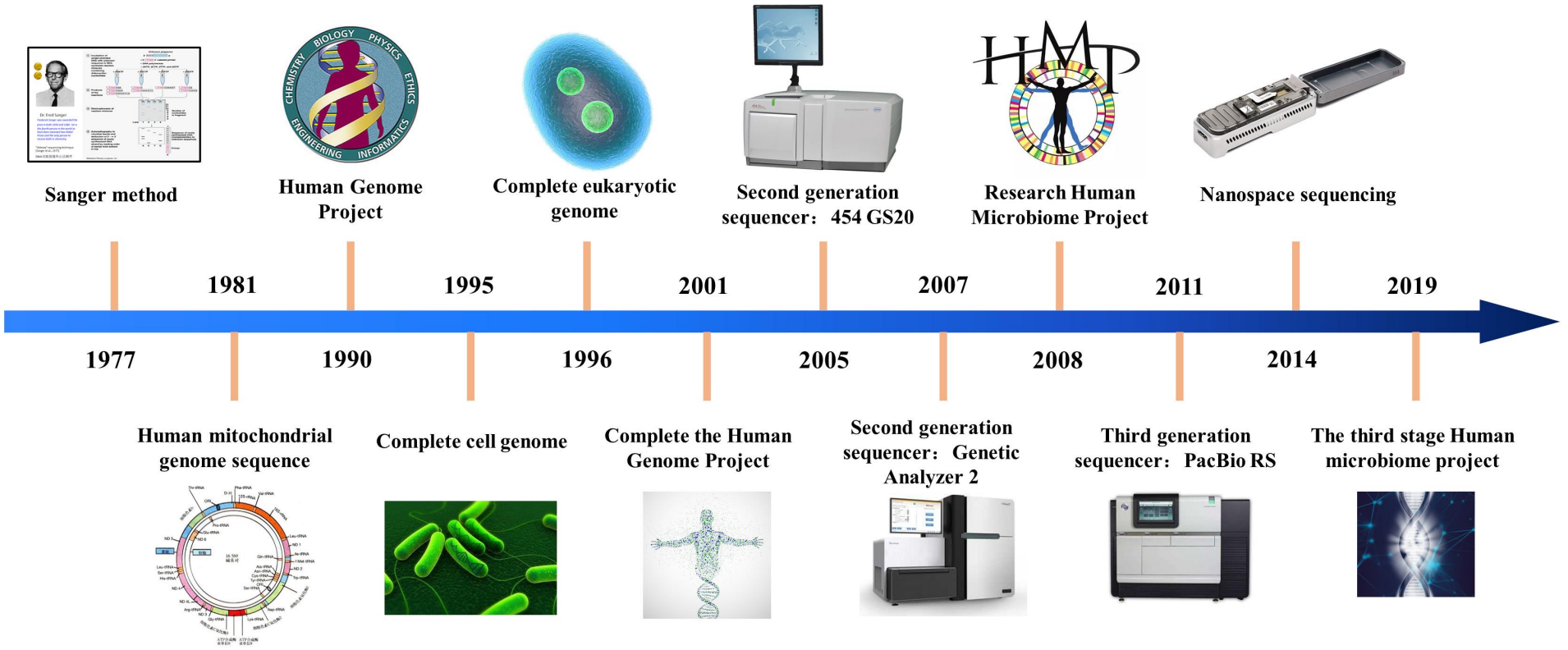
- **All exercises and presentations can be found from:**

  https://github.com/karkman/MMB-127_Genomics

- We will start every day at 10.00 with an introductory lecture and then hands-on exercises for the rest of the day

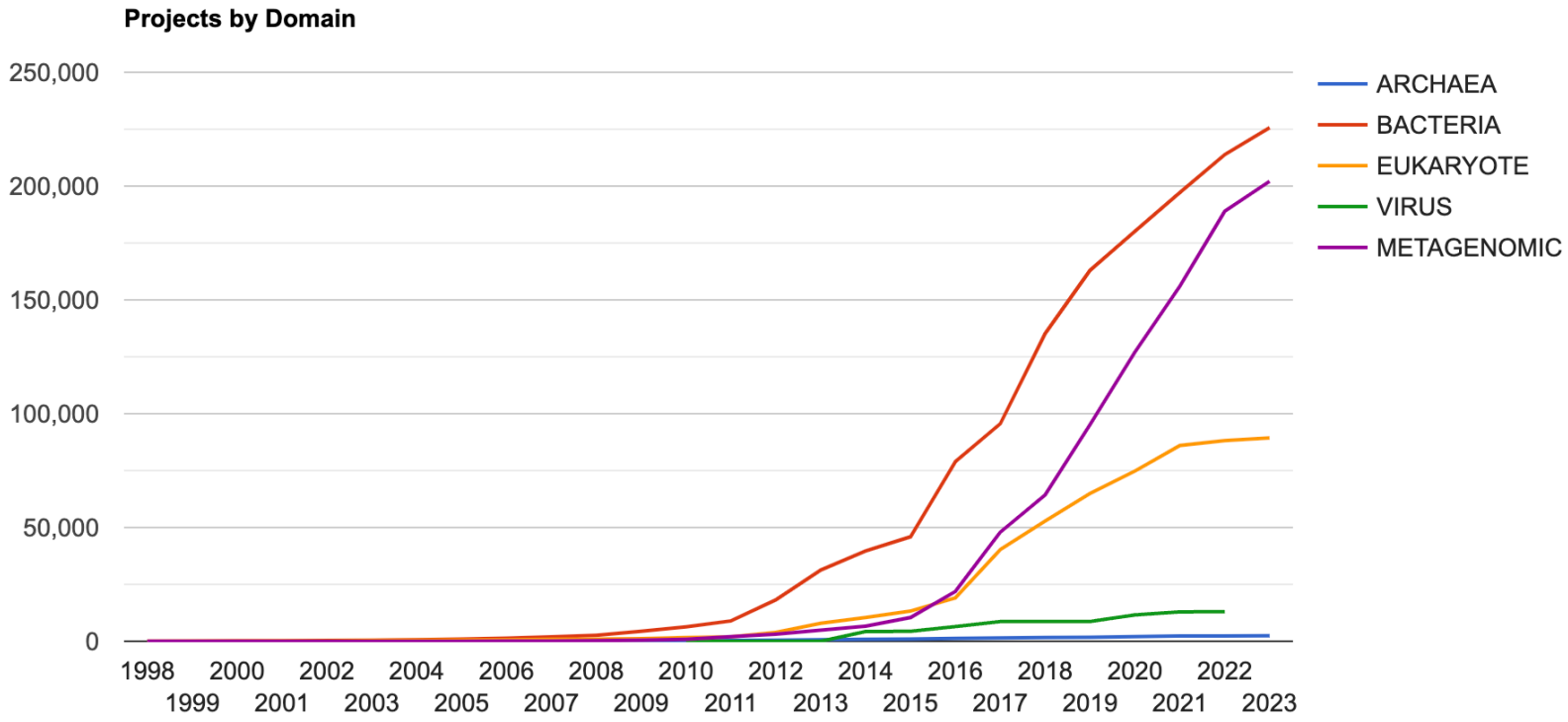
Whole genome sequencing (WGS)

The evolution of sequencing



https://doi.org/10.3389/fbioe.2020.01032

Genomes OnLine (GOLD) database

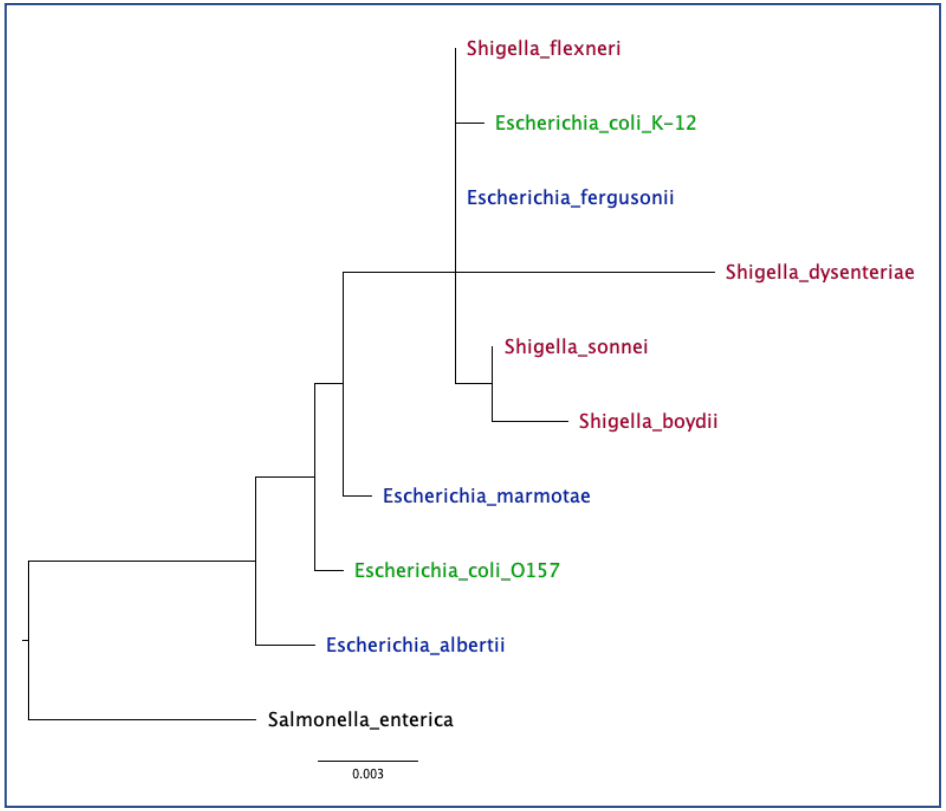


https://gold.jgi.doe.gov/statistics

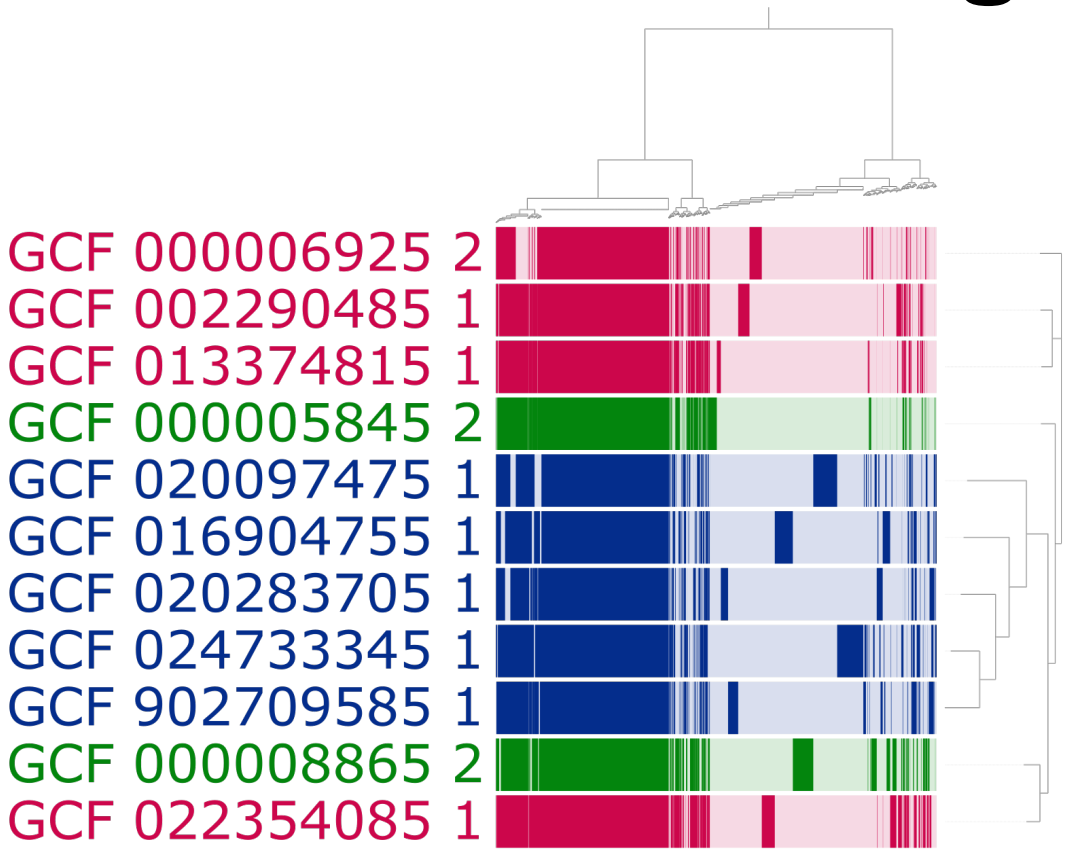
Why whole genome sequencing?

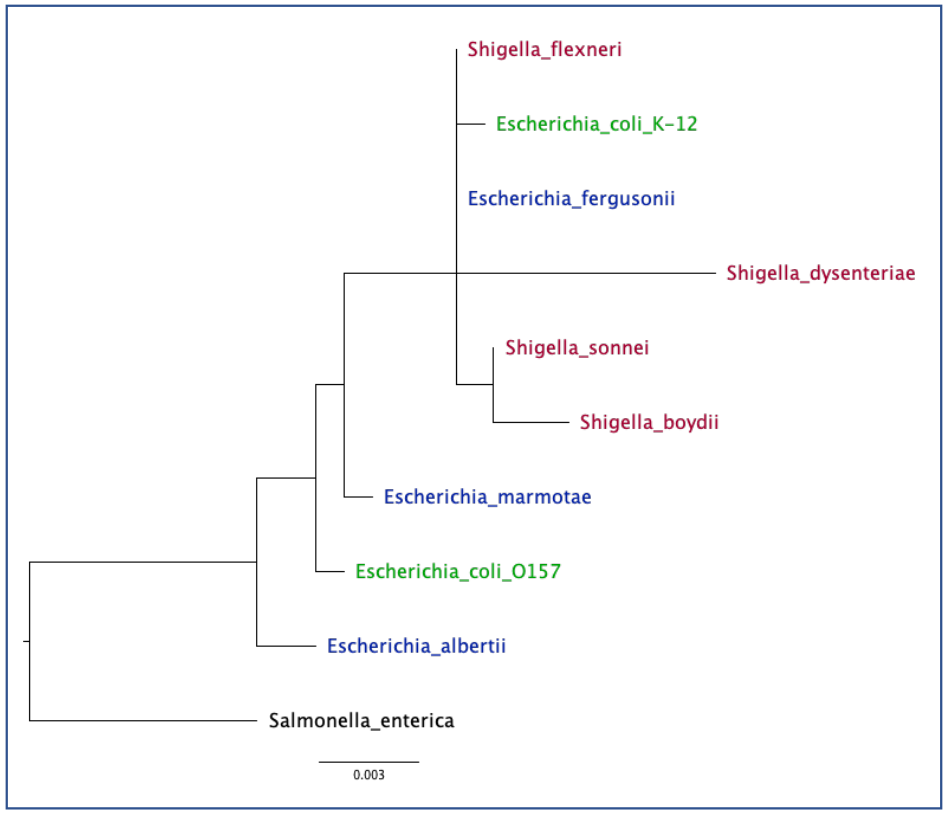
- *Escherichia* and *Shigella* as an example
- 16S rRNA gene cannot resolve the two genera
  (or these belong to the same genus)

- **What can you tell about your bacteria based on the 16S rRNA gene?**



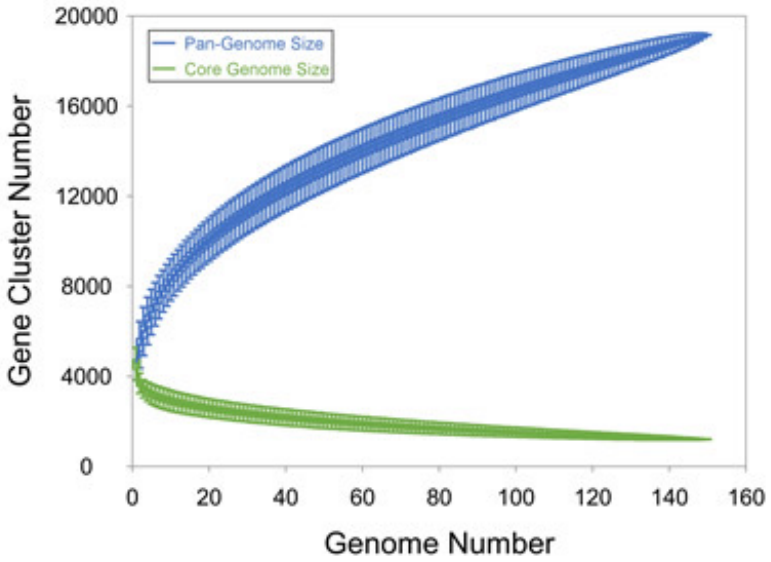
Escherichia–Shigella WGS vs. 16S



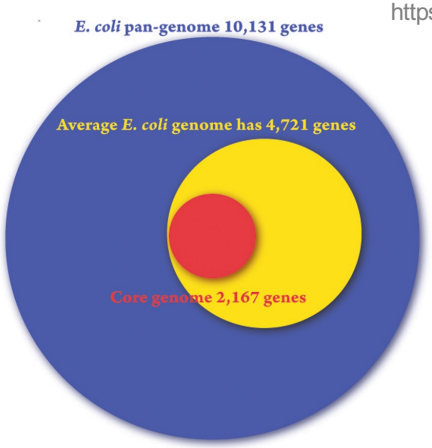


Pangenomes

- Bacterial species can have very diverse genomes
- Bacterial lifestyle affects pangenome size
- **Pangenome:** All genes from the studies strains
- **Core-genome:** genes shared by all strains
- **Accessory genome:** genes shared by only one or few strains



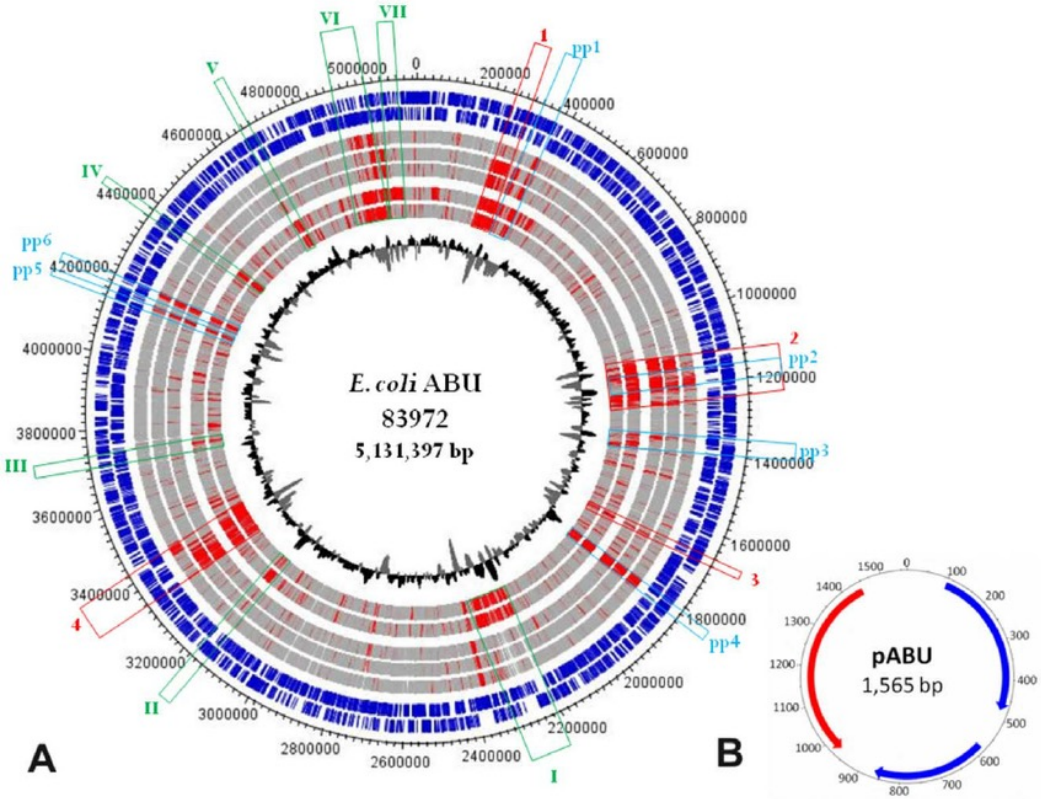
https://pangp.zhaopage.com/



https://doi.org/10.1371/journal.pgen.1000335

Genomics can be used to study

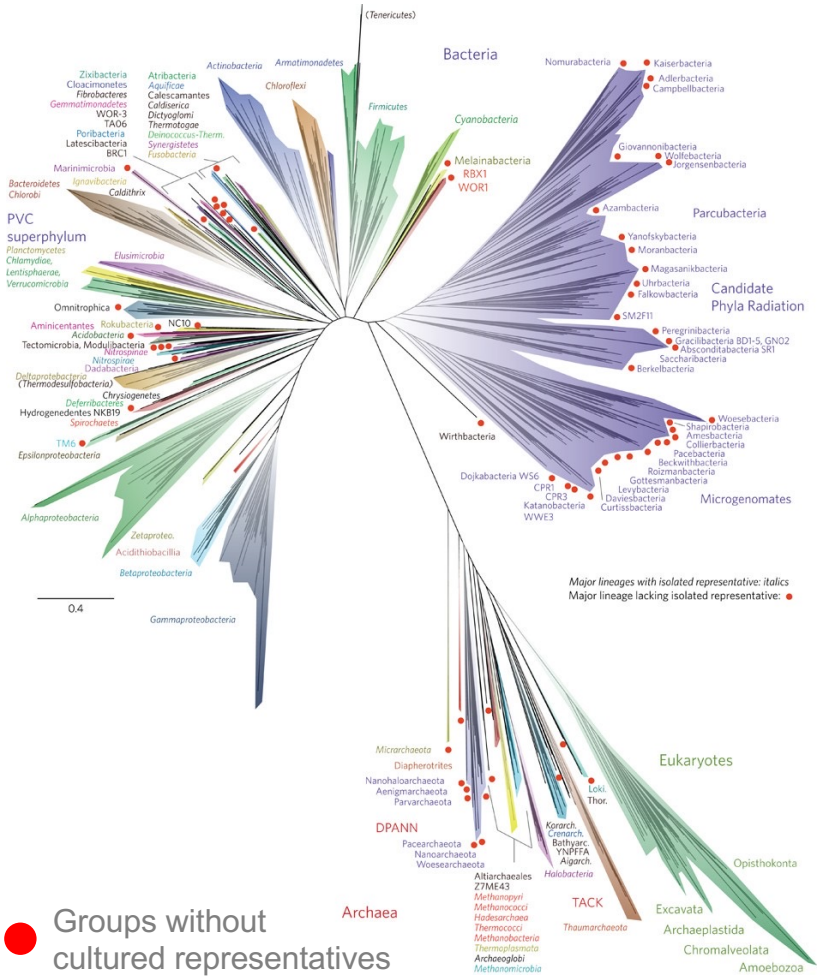
- Physiology
- Evolution
- Pathogenesis
- Novel industrial processes
- Novel diagnostic/ epidemiologic tests
- Extra-chromosomal elements



https://doi.org/10.1371/journal.ppat.1001078

The new tree of life

- Isolation of most microorganisms from the environment is not trivial
- Recent metagenomic assessments suggest the existence of up to 1 trillion microbial species in our planet
- But < 10,000 bacterial species have been described so far



https://doi.org/10.1038/nmicrobiol.2016.48

Your strains

What was done by you in the lab?

- Isolation of the strain
- DNA extraction
- DNA measurements
  - Purity, integrity, contamination
- Nanopore sequencing

What was done at the sequencing lab?

- Quality control (size, amount)
- Library preparation with the Nextera kit
- Sequencing with Illumina MiSeq

Illumina library preparation

Size selection

Correct amount of input DNA to avoid under- and overtagmentation

Adapter ligation

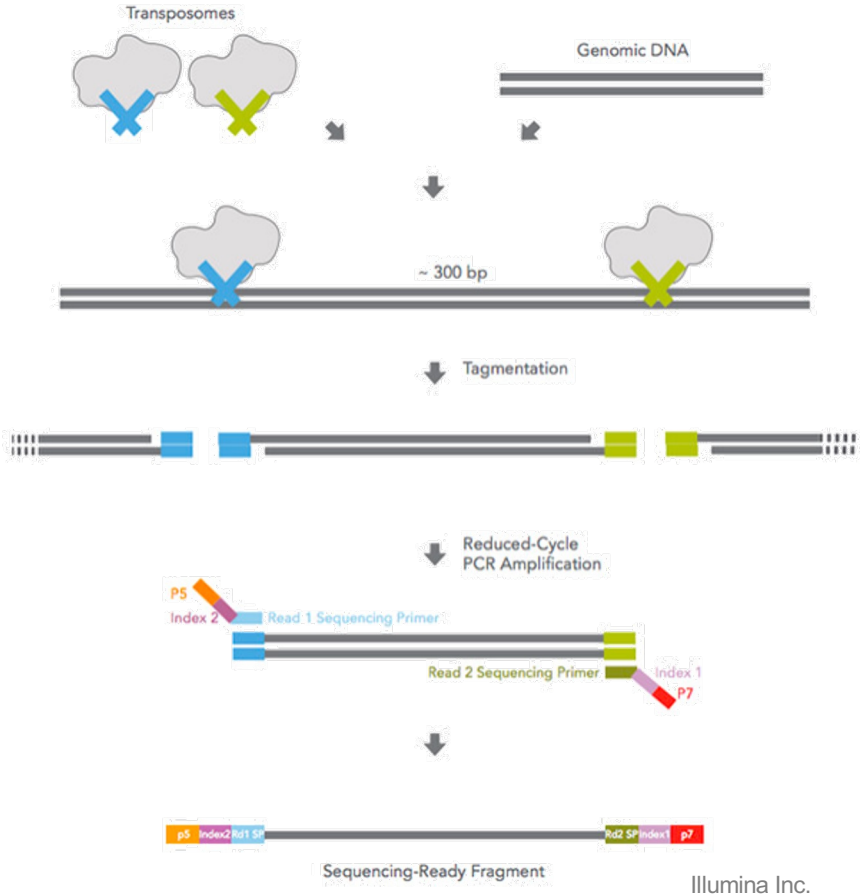
Flow cell adapters

Sequencing primers

Sequencing indexes

Optional indexes for multiplexing

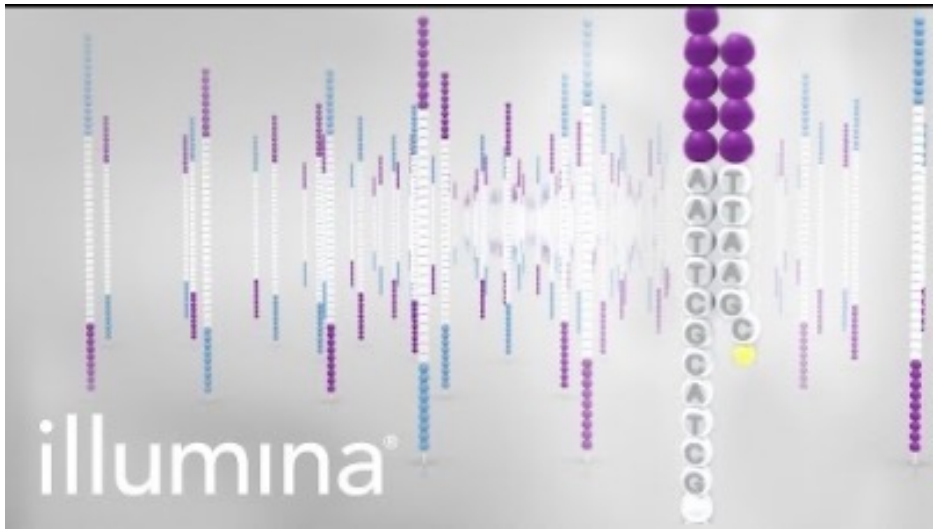
Biases in the first bases have been observed



Illumina Inc.

Illumina sequencing

- Good compromise between size, amount and error rate of reads
- The longer the reads the better.
- Current long-read technologies are becoming more relevant due to lower price and better sequence quality



https://www.youtube.com/watch?v=fCd6B5HRaZ8

Bioinformatics

- Interdisciplinary field that develops and applies computational methods to analyse biological data such as DNA and protein sequences

- Biology meets information science

- **A rapidly-evolving field**
  - Software development
  - Amount and type of data generated

UNIX and CSC

UNIX

- Family of computer operating systems (OS)
  - Linux, macOS, Solaris, OpenBSD

- Key characteristics
  - Multitasking
  - Multiuser
  - Multiprocessing
  - Portable

The UNIX philosophy

“The idea that the power of a system comes more from the relationships among programs than from the programs themselves”

- Use of plain text for storing data
- Use of a hierarchical filesystem
- Use of a large number of simple programs performing a limited, well-defined function
- Use of a command-line interpreter (“shell”) to combine these programs to perform complex tasks

The UNIX shell

- Command-line interpreter
- Interprets sequences of text
- Entered by a user
- From a file
- From a data stream

- Primary interface before graphical user interfaces (GUIs) appeared

- Still widely used today
  - Efficient
  - Low memory footprint
  - Advanced scripting



Some basic UNIX commands

pwd	print working directory (“where am I?”)
ls	list (“show folder contents”)
mkdir	make directory (a.k.a. folder)
cd	change directory (“go to folder”)
cp	copy
mv	move
rm	remove

Some additional notes

Case-sensitive

`photo.jpg ≠ PHOTO.jpg`

Does not like spaces and special characters in file/folder names

`genome report.txt` ❌
`genome_report.txt` ✅
`väitöskirja.txt` ❌
`vaitoskirja.txt` ✅

Space after each “word” in the command

Commands have to be typed in a single line, one at a time

After each command, hit “Enter” to execute it

Lines starting with “#” are comments

A few tricks:

- Tabulator (the key)
- History (up arrow)

How to learn UNIX?

- Bu using it!
  - Trial and error
  - Don’t copy/paste, type yourself
- Ask the internet
  - http://stackoverflow.com/
  - http://stackexchange.com/
  - http://askubuntu.com/
  - Search engines
  - **AI**
- Online courses/tutorials
  - http://codecademy.com
- Cheat sheets
  - https://www.stationx.net/unix-commands-cheat-sheet/
- Manual pages
  - `man <program>`

CSC - IT CENTER FOR SCIENCE

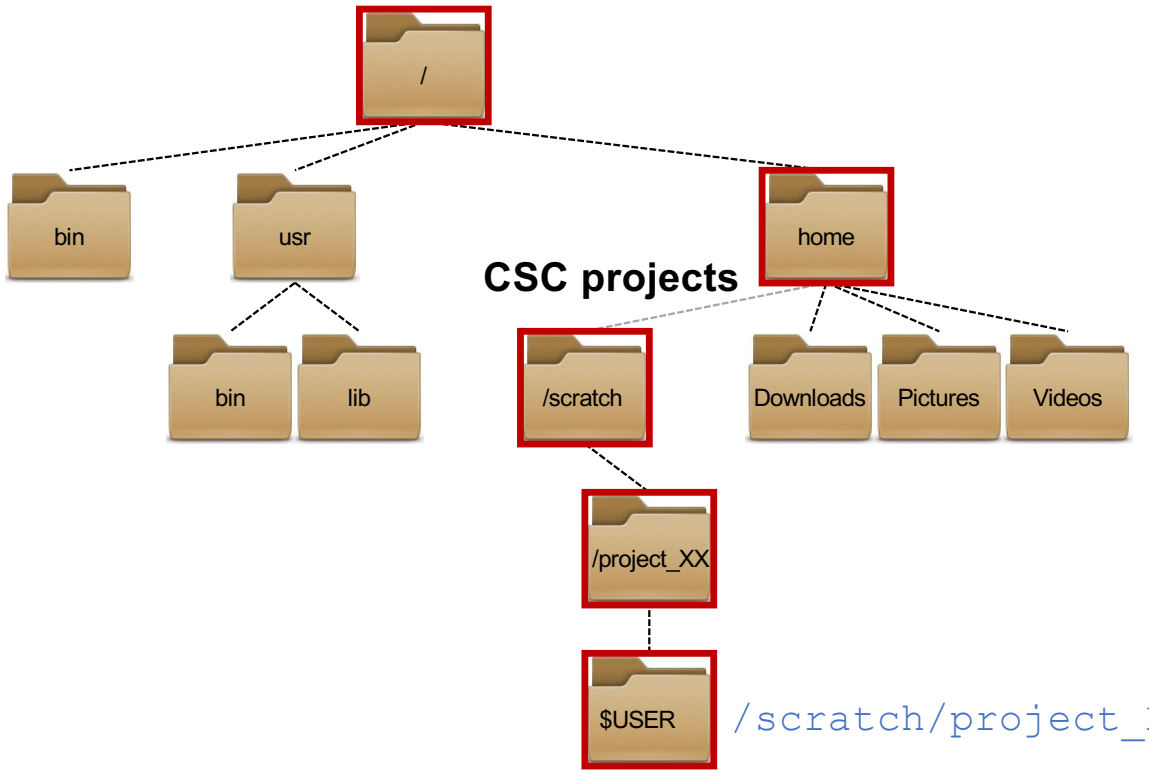
- ”Non-profit state enterprise with special tasks”
- Provides computing services for academics, research institutes and companies
- Free for academic research
- Owned by Finnish state (70 %) and higher education institutions (30 %)
- Funding from the Ministry of education and culture

Roihu

- One of the supercomputers at CSC
- Most used for bioinformatics
- Has (a lot of) CPU and GPU nodes
- Large collection of pre-installed software (modules)
- Interactive use and batch job scheduling system (SLURM)

- Read more: https://docs.csc.fi/computing/systems-roihu/

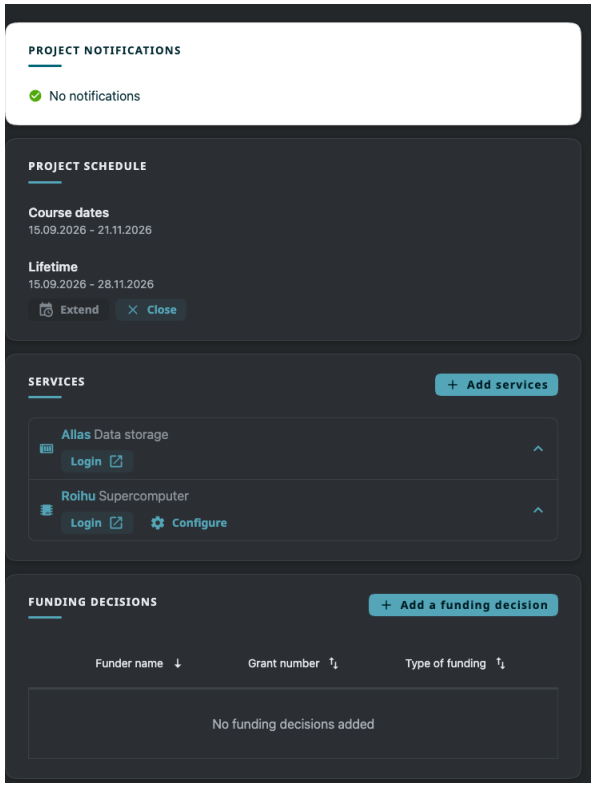
The filesystem in Puhti



`/scratch/project_XXXXXX/$USER`

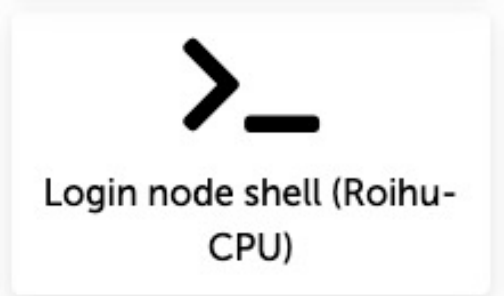
Before we start, check if everything is OK in CSC

- Login to https://my.csc.fi/
- Go to **Projects**
- Click on **MMB-127**
- Check that you can see Roihu on the right side under “Services”
- If you want to use SSH connection from your own laptop:
  - Make sure you know your CSC username and password



Connecting to Roihu

- Go to www.roihu.csc.fi
- Log in with Haka or CSC
- Select “Login node shell (Roihu-CPU)”
- Opens the command line on login nodes



- For computing select the “Compute node shell”



UNIX exercises

https://github.com/karkman/MMB-127_Genomics

Day 1: UNIX and CSC